

```
%chk=Pbaqua.chk
%mem=250GB
%nprocshared=50
# opt=loose freq apfd/gen scrf=(solvent=water) pseudo=read scf=xqc
```

Pbaqua

```
2 1
Pb      -0.01811500    0.00229600   -0.38950300
O        2.64998400   -0.51158300   -0.56189100
H        3.06040800   -0.42774300   -1.43398200
O        0.63138900   -1.85028000    1.13621000
H        0.04688300   -2.58656600    1.41100200
O        1.38004000    0.76260900    1.52258900
H        1.06698000    1.33136400    2.36838500
O       -0.71142000    2.55234200   -0.21417800
H       -1.16512200    2.91489600   -1.05275700
O       -1.45058700    0.53099700    1.54643700
H       -2.21614400   -0.06066900    1.51356200
O       -2.32898300   -1.50247100   -0.23186800
H       -2.30132000   -2.46569100   -0.27433000
H       -3.00544600   -1.24686700   -0.86801400
H       -0.00503300    3.23687800    0.10752900
H       -1.77628100    1.45670400    1.34354000
H        2.31625600    1.02522500    1.27342400
H        1.01424400   -1.43953000    1.97280000
H        2.80664100   -1.41917000   -0.29033200
```

```
H O 0
6-311+g(2d,p)
****
```

```
Pb      0
S      4  1.00
      591.61124370      0.22126521076D-03
      46.757232559      0.56961959130D-02
      20.746462696     -0.21374063831
      14.610796419      0.40502620616
S      2  1.00
      20.222637612     -0.83541883299D-01
      6.4767324865      0.97910892388
S      1  1.00
      1.6600600927      1.0000000
S      1  1.00
      0.80431655001      1.0000000
S      1  1.00
      0.22627039020      1.0000000
S      1  1.00
      0.84014530665D-01      1.0000000
P      3  1.00
      15.189102118      0.61952303583
```

		14.693144415	-0.72498497086
		6.8705890048	0.37680007984
P	3	1.00	
		2.2021426123	0.40196284806
		1.2209125119	0.46058131862
		0.63367559815	0.19367655397
P	1	1.00	
		0.28202837058	1.00000000
P	1	1.00	
		0.11333375666	1.00000000
P	1	1.00	
		0.43948707430D-01	1.00000000
D	6	1.00	
		61.315369628	0.33870800787D-03
		12.372195840	0.13788683942D-01
		6.9254944983	-0.75979608103D-01
		2.3319539939	0.28113784298
		1.2108730003	0.44474512269
		0.60090478506	0.35326874351
D	1	1.00	
		0.28135869813	1.00000000
D	1	1.00	
		0.115000000000	1.00000000
F	1	1.00	
		1.00000000	1.00000000
F	1	1.00	
		0.2896200	1.00000000

PB 0

PB-ECP 3 60

f potential

2

2	3.88751200	12.20989200
2	3.81196300	16.19029100

s-f potential

4

2	12.29630300	281.28549900
2	8.63263400	62.52021700
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

p-f potential

6

2	10.24179000	72.27689700
2	8.92417600	144.59108300
2	6.58134200	4.75869300
2	6.25540300	9.94062100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

d-f potential

6

2	7.75433600	35.84850700
2	7.72028100	53.72434200
2	4.97026400	10.11525600
2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100